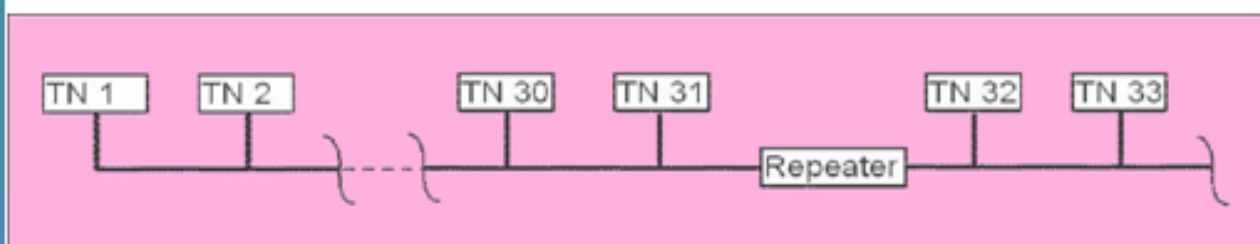


PROFIBUS

Installation Guideline for Planning

Supplement



Version 1.0

August 2009



PROFIBUS

Installation Guideline for Planning

Supplement

Version 1.0

August 2009

Order No: 8.042

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Revision log

Version	Date	Modifications/History
Version 1.0	August 18, 2009	Initial Release

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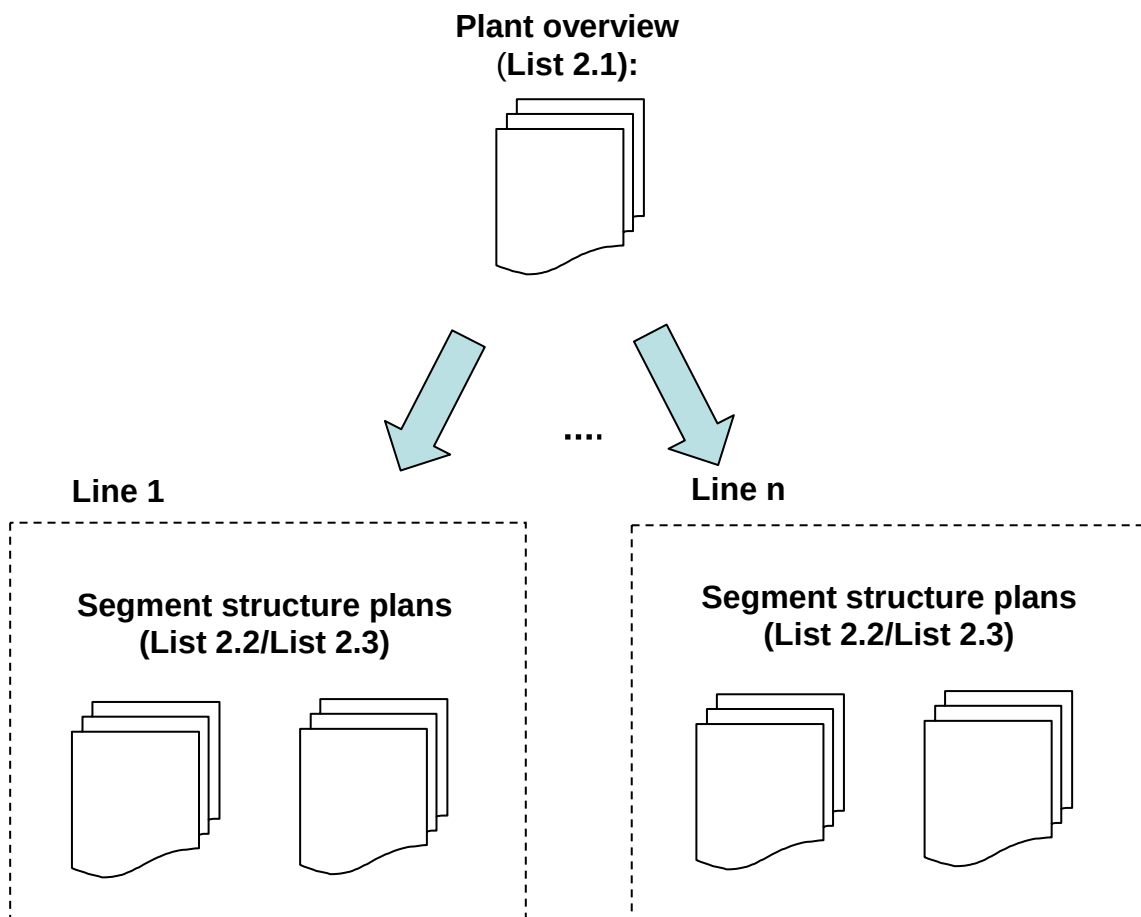
- **List 5.1:** Checklist for chapter 2
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1. Introduction

This supplement has been compiled as an amendment to the PROFIBUS Design Guideline. Its major purpose is to provide separate reference lists, material lists, support lists and check lists which need to be completed when working through this guideline. These lists are templates which may have to be copied several times depending on the scope of the automation project. They are provided to support the planning staff while not claiming to be exhaustive. Before using these lists, you should check whether additional details may have to be added for your specific system.

2. Reference lists

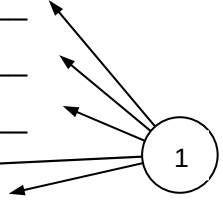
- **List 2.1:** Plant overview
- **List 2.2:** Segment structure plan (PROFIBUS DP/DP-IS)
- **List 2.3:** Segment structure plan (PROFIBUS-MBP (PA))

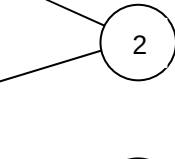
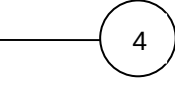
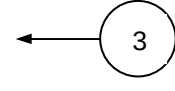


Plant overview

Automation unit: Exemplary unit
Controller: Controller
Number of lines: 4

Line No.: 1 Plant part(s): Exemplary plant



No. of PROFIBUS DP/DP-IS segments	<u>1</u>	
Number of bus nodes	<u>5</u>	
Data transmission rate	<u>1500 kbps</u>	
Communication protocol	<u>DP-V1</u>	
No. of PROFIBUS-MBP (PA) segments	<u>2</u>	
Number of bus nodes	<u>14</u>	

- 
-  Automation unit; controller; number of lines; line number(s); plant part(s)
 -  Data transmission rate / number of PROFIBUS DP segments (of line)
 -  Number of PROFIBUS-MBP (PA) segments (of line)
 -  Communication protocol

Plant overview

Automation unit: _____

Controller: _____

Number of lines: _____

Line No.: _____

Plant part(s): _____

No. of PROFIBUS DP/DP-IS segments

Number of bus nodes

Data transmission rate

Communication protocol

No. of PROFIBUS-MBP (PA) segments

Number of bus nodes

Line No.: _____

Plant part(s): _____

No. of PROFIBUS DP/DP-IS segments

Number of bus nodes

Data transmission rate

Communication protocol

No. of PROFIBUS-MBP (PA) segments

Number of bus nodes

Sheet _____ of _____

Segment structure plan

PROFIBUS DP/DP-IS

Automation unit:

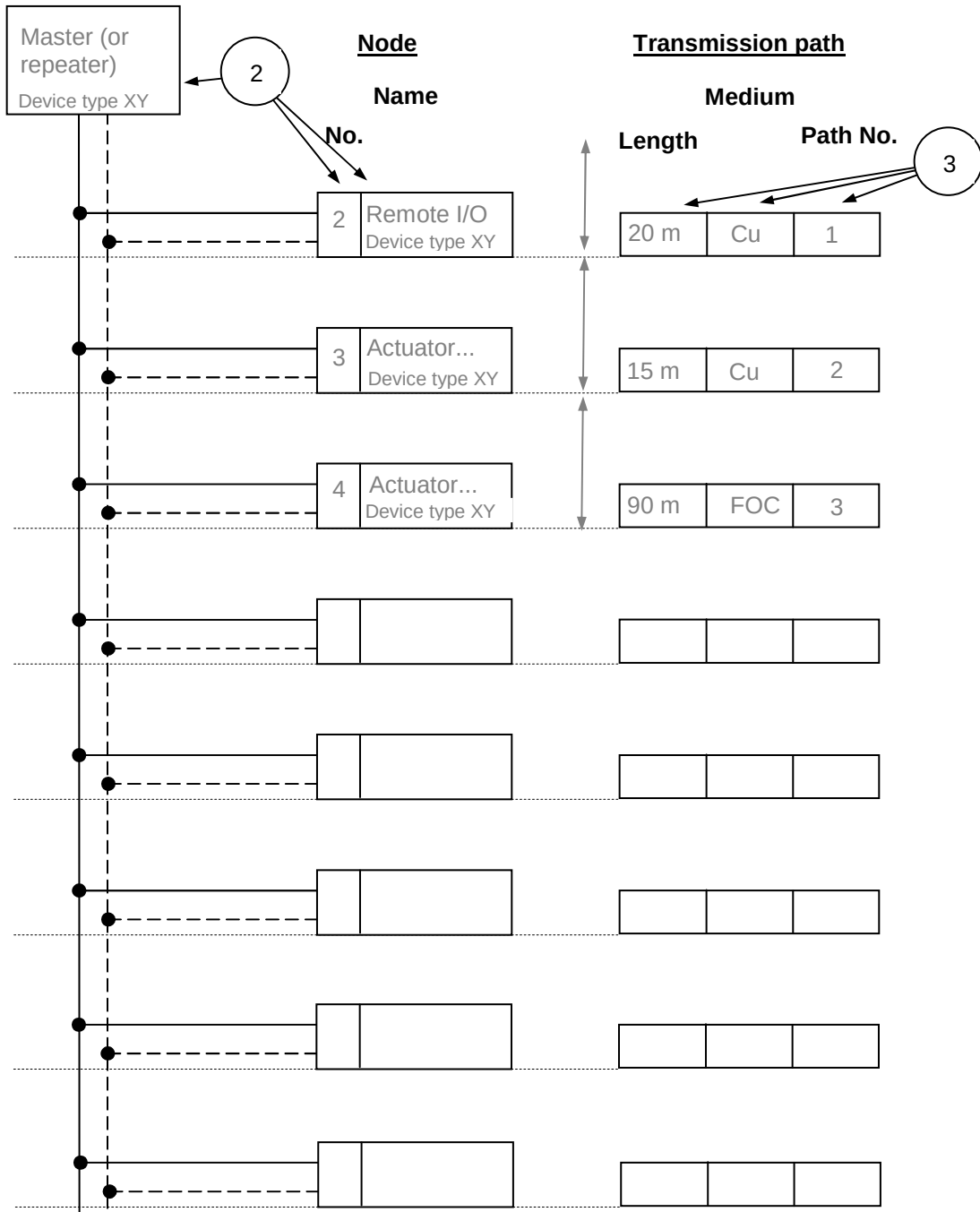
Plant part:

Exemplary unit

Exemplary unit

Line No.: 1

Segment No. 1 of 2



Segment structure plan

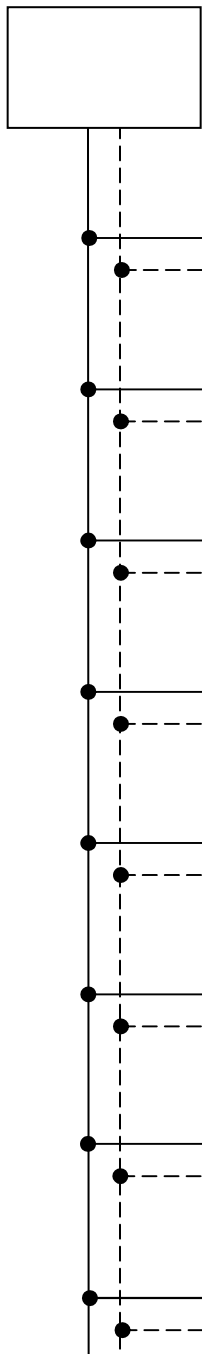
PROFIBUS DP/DP-IS

Automation unit: _____

Plant part : _____

Line No.: _____

Segment No. _____ of _____

	<u>Node</u>		<u>Transmission path</u>	
	Name		Medium	
	No.		Length	Path No.
				

Segment structure plan

PROFIBUS DP/DP-IS

Automation unit: _____

Plant part: _____

Line No.: _____

Segment No. _____ of _____

<u>Node</u>		<u>Transmission path</u>	
Name		Medium	
No.		Length	Path No.
•			
•			
•			
•			
•			
•			
•			
•			
•			

Sheet _____ of _____

Segment structure plan PROFIBUS-MBP (PA)

Automation unit:

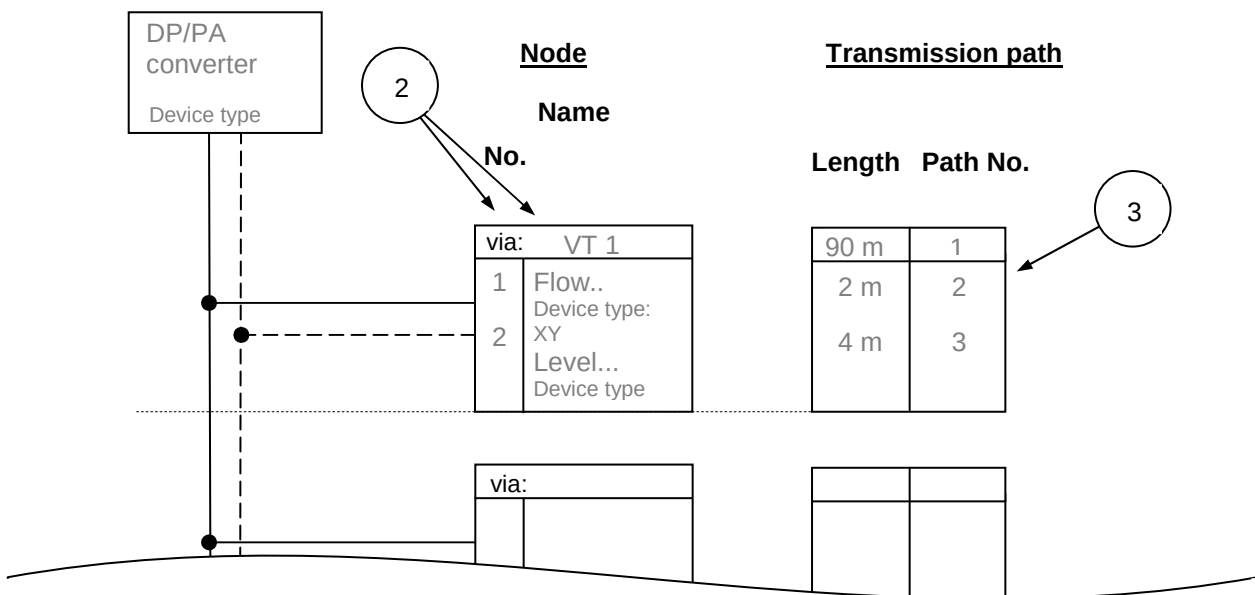
Plant part:

Exemplary

Exemplary

Line No.: 1

Segment No. 2 of 2



- 1 Automation unit; plant part; line and segment numbers
- 2 The connected bus nodes (node number; device type / name)
- 3 The approximate cable lengths of the main bus and the stubs (including path numbers) as well as the required junctions

Junction:		
VT1	Junction module	2 branches

Segment structure plan

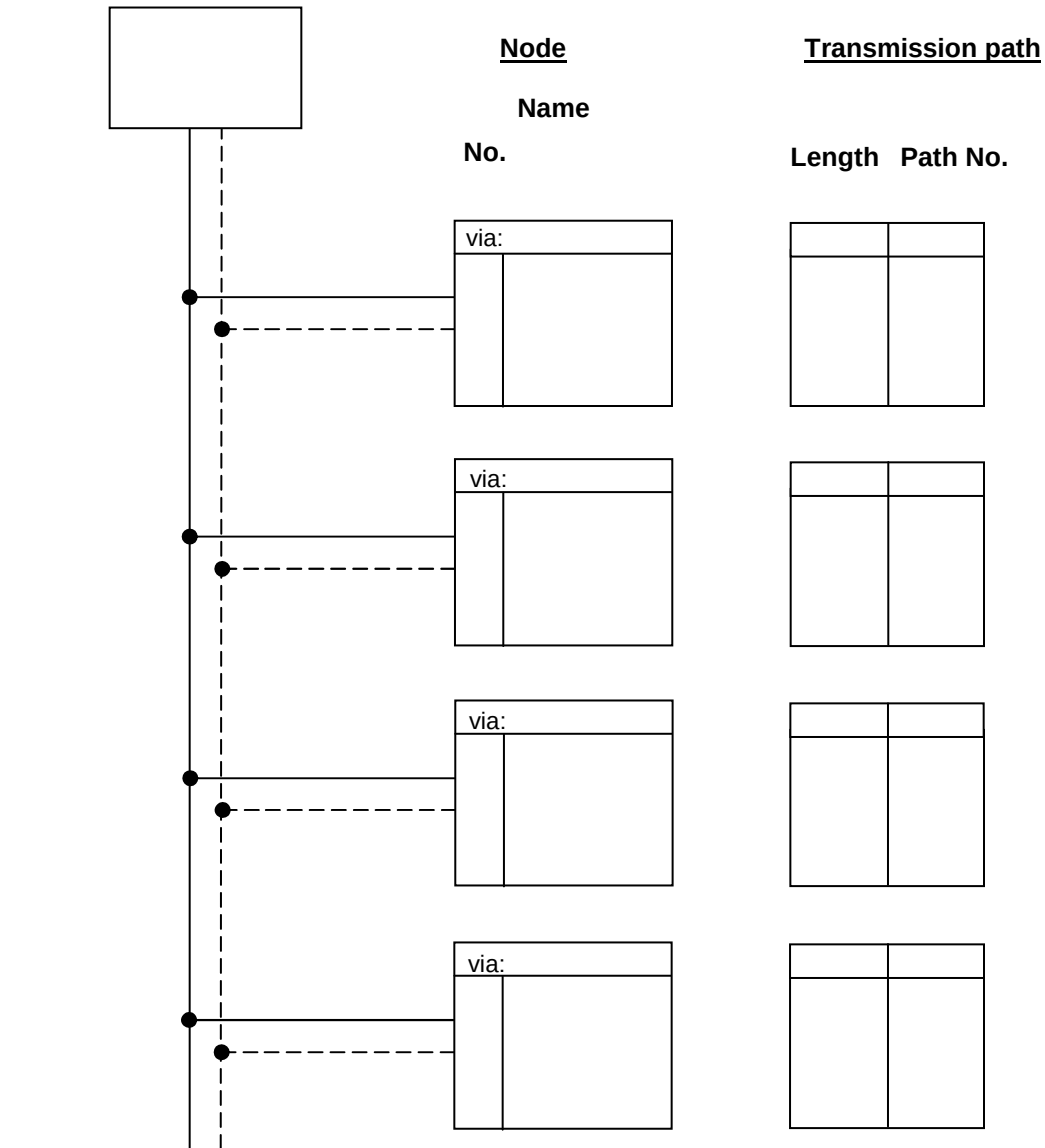
PROFIBUS-MBP (PA)

Automation unit: _____

Plant part: _____

Line No.: _____

Segment No. _____ of _____



Junction:		

Sheet _____ of _____

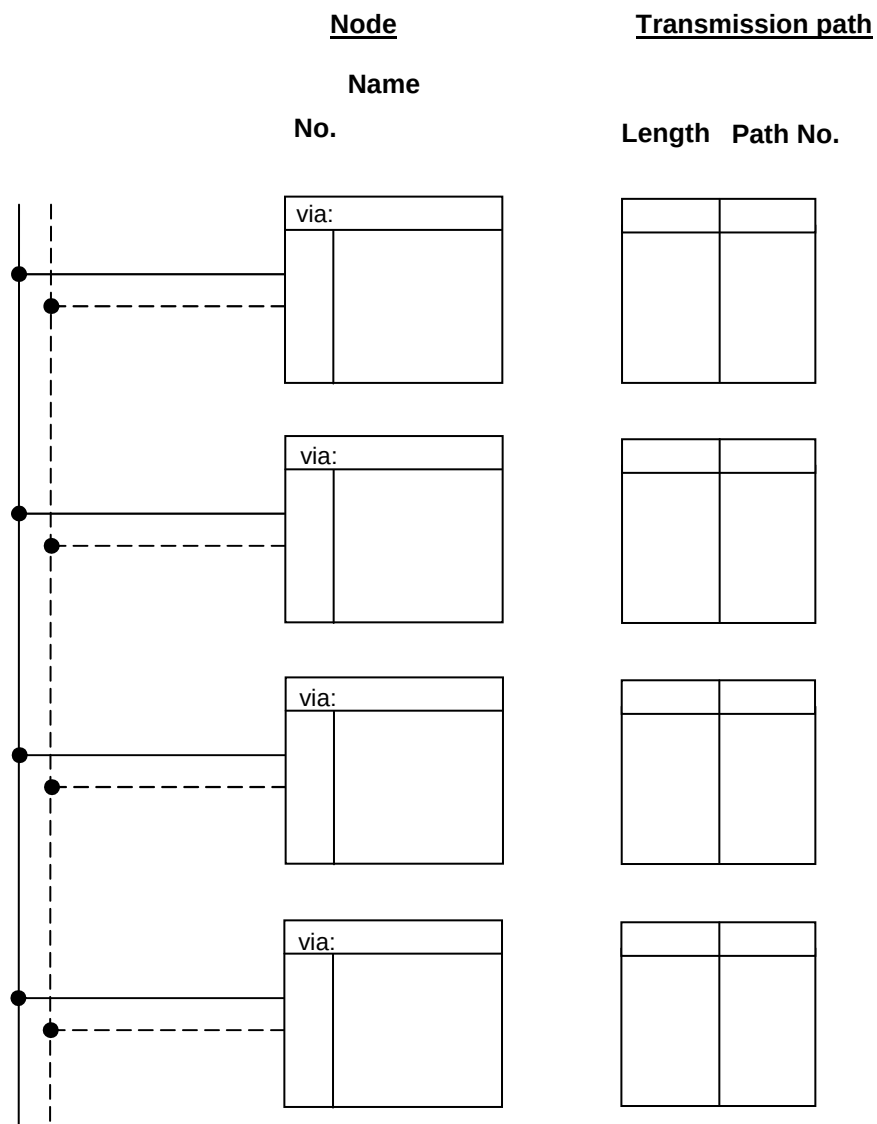
Segment structure plan PROFIBUS-MBP (PA)

Automation unit:

Plant part:

Line No.: _____

Segment No. _____ of _____



Junction:		

Sheet _____ of _____

3. Material lists

Material lists for PROFIBUS DP/DP-IS

- **List 3.1:** Cables and connections overview
- **List 3.2:** Cable material list
- **List 3.3:** Connector material list
- **List 3.4:** Repeater material list

Automation unit: _____

[illegible]

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Cable material list

PROFIBUS DP/DP-IS cables

Automation unit: _____

Cable material no.: _____

Design type	
Sheath color	
Cable type/fiber type	
Cable diameter	
Wire configuration	
Notes / special features	

Required cables

Line No.: _____

Customizable cable	Ready-made cable				
Length	<table style="width: 100%;"><tr><td style="width: 60%;">Length</td><td style="width: 40%;">Connectors</td></tr><tr><td colspan="2" style="height: 150px; border-left: 1px solid black; border-right: 1px solid black;"></td></tr></table>	Length	Connectors		
Length	Connectors				

Line No.: _____

Customizable cable	Ready-made cable				
Length	<table style="width: 100%;"><tr><td style="width: 60%;">Length</td><td style="width: 40%;">Connectors</td></tr><tr><td colspan="2" style="height: 150px; border-left: 1px solid black; border-right: 1px solid black;"></td></tr></table>	Length	Connectors		
Length	Connectors				

Total length: _____

Sheet _____ of _____

Connector material list

PROFIBUS DP/DP-IS connectors

Automation unit: _____

Design type	
Notes / special features	

Required connectors

Line No.: _____

Required quantity	
-------------------	--

Line No.: _____

Required quantity	
-------------------	--

Line No.: _____

Required quantity	
-------------------	--

Line No.: _____

Required quantity	
-------------------	--

Total number: _____

Sheet _____ of _____

Repeater material list

PROFIBUS DP

Automation unit: _____

Required repeaters

Line No.: _____

Name/type	
Regenerating	
Segments	
Terminating resistors	

Line No.: _____

Name/type	
Regenerating	
Segments	
Terminating resistors	

Line No.: _____

Name/type	
Regenerating	
Segments	
Terminating resistors	

Total number: _____

Sheet _____ of _____

Material lists for PROFIBUS-MBP (PA)

- **List 3.5:** Cables and connections overview
- **List 3.6:** Cable material list
- **List 3.7:** Connector material list
- **List 3.8:** Junction material list
- **List 3.9:** Bus termination material list

Automation unit: _____

[illegible]

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Cable material list

PROFIBUS-MBP (PA) cables

Automation unit: _____

Cable material No.: _____

Design type	
Sheath color	
Line type	
Cable diameter	
Wire configuration	
Notes / special features	

Required cables

Line No.: _____

Customizable cable	Ready-made cable
	LengthConnectors

Line No.: _____

Customizable cable	Ready-made cable
Length	LengthConnectors

Total length: _____

Sheet _____ of _____

Connector material list

PROFIBUS-MBP (PA) connectors

Automation unit: _____

Design type	
Notes / special features	

Required connectors

Line No.: _____

Required quantity	
-------------------	--

Line No.: _____

Required quantity	
-------------------	--

Line No.: _____

Required quantity	
-------------------	--

Line No.: _____

Required quantity	
-------------------	--

Total number: _____

Sheet _____ of _____

Junction material overview

PROFIBUS-MBP (PA) junctions

Automation unit: _____

Design type	
Notes / special features	

Required junctions

Line No.: _____

Required quantity	
-------------------	--

Line No.: _____

Required quantity	
-------------------	--

Line No.: _____

Required quantity	
-------------------	--

Line No.: _____

Required quantity	
-------------------	--

Total number: _____

Sheet _____ of _____

Bus termination material list

PROFIBUS-MBP (PA) bus terminations

Automation unit: _____

Design type	
Notes / special features	

Required bus terminations

Line No.: _____

Required quantity	
-------------------	--

Line No.: _____

Required quantity	
-------------------	--

Line No.: _____

Required quantity	
-------------------	--

Line No.: _____

Required quantity	
-------------------	--

Total number: _____

Sheet _____ of _____

4. Support lists

- **List 4.1:** Verification of a PROFIBUS DP-IS segment
- **List 4.2:** Verification of a PROFIBUS-MBP (PA) segment
- **List 4.3:** Verification of a PROFIBUS-MBP (PA) segment according to FISCO
- **List 4.4:** Verification of the cable routing

Verification of a PROFIBUS DP-IS segment

Automation unit: _____

Line No.: _____ Segment No.: _____

Verification of a PROFIBUS DP-IS segment			
Node	Approval for Ex area	Operation with DP-IS	Output power
Bus termination 1			
Bus termination 2			
Total power			
Meets criteria for intrinsic safety			

Sheet _____ of _____

Verification of a PROFIBUS-MBP (PA) segment

Automation unit: _____

Line No.: _____ Segment No.: _____

Verification of a PROFIBUS-MBP (PA) segment			
Slave	Approval for Ex area	FISCO approval	Power requirement
Total power requirements			
Segment designed according to FISCO			

Sheet _____ of _____

Verification of PROFIBUS-MBP (PA) segment according to FISCO model (page 1 of 2)

Automation unit: _____

Line No.: _____ Segment No. _____

Cables used				
Cable material No.:				
Cable parameters		Requirements	Requirements met	
			Yes	No
Resistance load per unit length		$15\Omega/\text{km} < R < 150\Omega/\text{km}$		
Inductance per unit length		$0,4\text{mH}/\text{km} < L < 1\text{mH}/\text{km}$		
Capacitance per unit length		$45\text{nF}/\text{km} < L < 200\text{nF}/\text{km}$		

Line lengths				
		Requirements	Requirements met	
			Yes	No
Line length		$\leq 1000\text{m}$ for EEx ia (IIC); $\leq 1900\text{m}$ for EEx ib (IIC/IIB)		

Stubs				
		Requirements (number / length)	Requirements met	
			Yes	No
Longest stub		1-18 / 60 m		
Number of stubs		19-24 / 30 m		
		25-32 / ---		

Sheet _____ of _____

Verification of a PROFIBUS-MBP (PA) segment according to FISCO model (page 2 of 2)

Power supply unit			
No.	Name	FISCO certified	
		Yes	No
1			

[illegible]

Bus terminations			
No.	Name	FISCO certified	
		Yes	No

Sheet _____ of _____

Verification of the cable routing

PROFIBUS DP / DP-IS / MBP (PA)

Automation unit: _____

Line No.: _____ Segment.: _____

	Path No.							
	1	2	3	4	5	6	7	n
Cable routing inside buildings								
Is the routing recorded in the floor plan?								
Is the cable routing separate from the power supply?								
Have metallic cable ducts been provided? Length?								
Have appropriate cut-off bridges been provided?								
For EMI: Steel pipes for shielding? Length?								
Has grounding of the cable ducts been provided?								
Are the cable ducts protected against corrosion?								
Have the bending radii been observed?								
Cable routing in the cabinet								
Has a grounding bar been provided?								
Have separate cable routings been provided?								
Are minimum distances to be kept?								
Have cut-off bridges been provided?								
Have rectangular cable crossings been realized?								
Have the cable bundles been combined correctly?								
Are bending radii to be observed?								
Cable routing outside buildings								
Are plastic pipes provided for protection?								
Are minimum distances to be kept?								
Has an intermediate terminal box been provided?								
Has a lightning protection been provided?								
Are bending radii to be observed?								
Mechanical protection								
Plastic pipes for laying off the route?								
Armored pipes for mechanical stress?								
Armored pipes provided for walkable areas?								
Damage excluded for bends?								
Are bending radii to be observed?								

Sheet _____ of _____

5. Checklists

- **List 5.1:** Checklist for chapter 2
- **List 5.2:** Checklist for chapter 3.1
- **List 5.3:** Checklist for chapter 3.2
- **List 5.4:** Checklist for chapter 3.3
- **List 5.5:** Checklist for chapter 3.4
- **List 5.6:** Checklist for chapter 4
- **List 5.7:** Checklist for chapter 5

Checklist for chapter 2 (list 5.1)

Determining the PROFIBUS structure (chapter 2.1)

		Yes	No
1.	Is the automation plant divided into separate automation units?	☐	☐
2.	Are the lines clearly assigned to automation units and plant parts?	☐	☐
3.	Are all devices supplied from one power supply (safety)?		
4.	Are the lines divided into segments?	☐	☐
5.	Are the segments assigned to a PROFIBUS type?	☐	☐
6.	Is the topology of all lines known?	☐	☐
7.	Is the approximate length of all transmission paths known ?	☐	☐
8.	Have you checked whether there are areas with EMI problems?		

Determining data relevant for the line (chapter 2.2)

		Yes	No
1.	Have the data transmission rates to be set for the lines been determined?	☐	☐
2.	Have the maximum stub lengths been observed for PROFIBUS-MBP (PA) ?	☐	☐
3.	Has a function stage of the communication protocol been assigned to every line?	☐	☐
4.	Any information or tools available for determining the bus cycle time?	☐	☐

Special requirements (chapter 2.3)

		Yes	No
1.	Has the plant been checked for the presence of hazardous areas ?	☐	☐
2.	Are the segments located in such hazardous areas assigned to a PROFIBUS type?	☐	☐
3.	Has the plant been checked for safety-related tasks?	☐	☐
4.	Has the use of reserve cables been considered?	☐	☐

Checklist for chapter 3.1 (list 5.2)

Selecting cables (chapter 3.1.1)

	Yes	No
1. Has a cable of a specific design been selected for each transmission path?	☐	☐
2. Are the cables of line type A?	☐	☐
3. Is the use of normal cables sufficient (i.e. no special cables required)?	☐	☐

Determining connections (chapter 3.1.2)

	Yes	No
1. Have the connections of all components been determined?	☐	☐
2. Has the assembly type of the cables been determined?	☐	☐
3. Has a connection for diagnosis or parameterizing devices been provided in each segment?	☐	☐
4. Have the connector types been determined?	☐	☐

Planning bus terminations (chapter 3.1.3)

	Yes	No
1. Has a connector with integrated bus termination been provided for the first or the last node in the segment?	☐	☐
2. Have externally supplied bus terminations been provided for nodes that must offer the possibility of being deactivated in live operation?	☐	☐
3. Has a terminating resistor been provided for a repeater at the beginning of a segment?	☐	☐

PROFIBUS DP-IS coupling (chapter 3.1.4)

	Yes	No
1. Have the fieldbus couplers required for coupling been provided?	☐	☐
2. Is the planned installation site of the fieldbus couplers located outside the hazardous area?	☐	☐
3. Have grounding and power supply of fieldbus barriers been provided according to the manufacturer specifications?	☐	☐
4. Have the PROFIBUS DP-IS segments been checked according to the above steps?	☐	☐

Planning repeaters (chapter 3.1.5)

	Yes	No
1. Are repeaters required to extend paths?	☐	☐
2. Are repeaters required as junctions?	☐	☐
3. Are the required repeaters included in the material lists?	☐	☐

Checklist for chapter 3.2 (list 5.3)

Planning DP/PA transitions (chapter 3.2.1)

	Yes	No
1. Has the power requirement of the segments been determined?	☐	☐
2. Has a power supply unit been (pre-)selected?	☐	☐
3. Has the number of slaves in the segment been adapted when using fieldbus barriers?	☐	☐
4. Have the cable lengths (stubs) been updated?	☐	☐
5. Has a DP/PA converter been selected?	☐	☐
6. Are the planned installation sites of the DP/PA converters located outside the hazardous area?	☐	☐

Selecting cables (chapter 3.2.2)

	Yes	No
1. Has a cable of a specific design been provided for each transmission path?	☐	☐
2. Are the cables of line type A?	☐	☐
3. Is the use of normal cables sufficient (i.e. no special cables are required)?	☐	☐

Determining connections (chapter 3.2.3)

	Yes	No
1. Have the connections of all components been determined?	☐	☐
2. Has the assembly type of the cables been determined?	☐	☐
3. Has a connection for diagnosis or parameterizing devices been provided in each segment?	☐	☐
4. Have the connector types been determined?	☐	☐

Determining the junction technology (chapter 3.2.4)

	Yes	No
1. Have all required junctions been provided?	☐	☐
2. Has the area of operation (hazardous area) been considered when selecting junctions?	☐	☐
3. Have free connections been provided at the junctions?	☐	☐

Planning bus terminations (chapter 3.2.5)

	Yes	No
1. Has each segment been provided with two bus terminations?	☐	☐
2. Has the area of operation (hazardous area) been considered when planning the bus terminations?	☐	☐

Checklist for chapter 3.3 (list 5.4)

Planning of cable routing for copper cables (chapter 3.3.1)

		Yes	No
1.	Has the topology been transferred to the prepared floor plan, observing the specifications?	☐	☐
2.	Have the minimum distances been observed?	☐	☐
3.	Have any measures been taken for mechanical protection?	☐	☐

Planning of grounding and shielding (chapter 3.3.2)

		Yes	No
1.	Have existing equipotential bonding systems been transferred to the floor plan?	☐	☐
2.	Has component grounding been provided according to the PROFIBUS configuration?	☐	☐
3.	Have any measures been taken to avoid potential differences?	☐	☐
4.	Have the issues for EMC-compatible cable routing been taken into account during planning?	☐	☐

Checklist for chapter 3.4 (list 5.5)

Optical connection technology (chapter 3.4.2)

	Yes	No
1. Have splice boxes been included in the planning?	☐	☐
2. Have bundle wire branches been avoided?	☐	☐

Selecting fibers and cables (chapters 3.4.3 and 3.4.4)

	Yes	No
1. Has a fiber type been selected for each transmission path?	☐	☐
2. Has a common cable been selected for parallel FO paths?	☐	☐
3. Have reserve fibers been planned for fiber optic cables?	☐	☐
4. Have already existing devices been checked for possible connections for the planned fibers?	☐	☐
5. Has a cable of a specific design been provided for each transmission path?	☐	☐

Determining connections (chapter 3.4.5)

	Yes	No
1. Have the connections of all components been determined?	☐	☐
2. Has the assembly type of the cables been determined?	☐	☐

Determining fiber-fiber couplings (chapter 3.4.6)

	Yes	No
1. Have all paths been checked for potential couplings?	☐	☐
2. Have possibilities for realization been determined for occurring couplings?	☐	☐

Checking FO paths (chapter 3.4.7)

	Yes	No
1. Have the planned basic optical paths been checked?	☐	☐
2. Can the planned paths be realized?	☐	☐

Checklist for chapter 4 (list 5.6)

Determining power supply (chapter 4)

		Yes	No
1.	Is there a five wire system for power supply?	☐	☐
2.	If not, can such a system be retrofitted later?	☐	☐
3.	Has a SELV low voltage supply been provided according to the requirements on the cable path?	☐	☐

Checklist for chapter 5 (list 5.7)

Planning plant acceptance (chapter 5)

		Yes	No
1.	Is qualified staff and the required measurement equipment available for the acceptance test?	☐	☐
2.	Has the time required for the individual steps been estimated and considered in the project plan?	☐	☐
3.	Has a visual inspection been carried out?	☐	☐
4.	Have the acceptance measurements been carried out?	☐	☐
5.	Has the functional test been carried out according to the specified individual steps?	☐	☐
6.	Have all steps been logged?	☐	☐

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